

STAT 429: Introduction to Time Series Analysis

Homework Two

Fall 2026, University of Illinois Urbana-Champaign

Total Points = 104; Maximum Score = 100

For problems involving data analysis, you are free to use inbuilt functions from libraries in R or Python. Attach code snippets in your solutions corresponding to each problem.

1. The file “AudioNote.mp3” contains sound from a musical note.
 - a) Plot the waveform of the audio signal and describe its main characteristics. **(3 points)**
 - b) To the waveform, fit the model

$$y_t = \beta_0 + \beta_1 \cos(2\pi ft) + \beta_2 \sin(2\pi ft) + \epsilon_t, \quad \epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

Report a point estimate of f along with an appropriate uncertainty interval. **(5 points)**

- c) What is your best estimate of the frequency, in Hertz, corresponding to the musical note from which this sound file was generated? **(2 points)**
 - d) Based on your frequency estimate, identify the musical note. **(2 points)**
2. The dataset `lynx.csv` gives the annual numbers of lynx trappings in Canada from 1821 through 1934.
 - a) Plot the periodogram of the data and comment on its notable features. **(3 points)**
 - b) Fit the model

$$Y_t = \beta_0 + \beta_1 \cos(2\pi ft) + \beta_2 \sin(2\pi ft) + \epsilon_t, \quad \epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2). \quad (1)$$

Provide a point estimate and a 95% uncertainty interval for the frequency f and the corresponding period of oscillation. **(5 points)**

- c) Provide point estimates and 95% uncertainty intervals for $\beta_0, \beta_1, \beta_2$, and σ . **(7 points)**
 - d) Comment on whether model (1) is a good model for this dataset. **(2 points)**

3. Download the Google Trends data for the United States for the query *mask*. Use a monthly time series beginning in January 2004 and ending in August 2026.

Fit the single change-point model

$$Y_t = \beta_0 + \beta_1 I\{t > c\} + \epsilon_t, \quad \epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2), \quad (2)$$

where $I\{t > c\}$ equals 1 if $t > c$ and equals 0 otherwise.

- a) Provide a point estimate and a 95% uncertainty interval for the change-point parameter c . Explain whether your answers make intuitive sense in the context of this dataset. **(5 points)**
- b) Generate $N = 1000$ samples of the parameters (c, β_0, β_1) from the posterior distribution, and plot the corresponding functions

$$t \mapsto \beta_0 + \beta_1 I\{t > c\}$$

along with the original data. **(5 points)**

- c) Comment on whether model (2) is a good model for this dataset. **(2 points)**

4. In Figure 1, you will find two different time-series datasets, Dataset One and Dataset Two, of the same length $n = 1000$. In Figure 2, you will find two periodograms, Periodogram A and Periodogram B.

One of the periodograms corresponds to Dataset One and the other corresponds to Dataset Two. Identify the correct matching, giving reasons for your answer. **(6 points)**

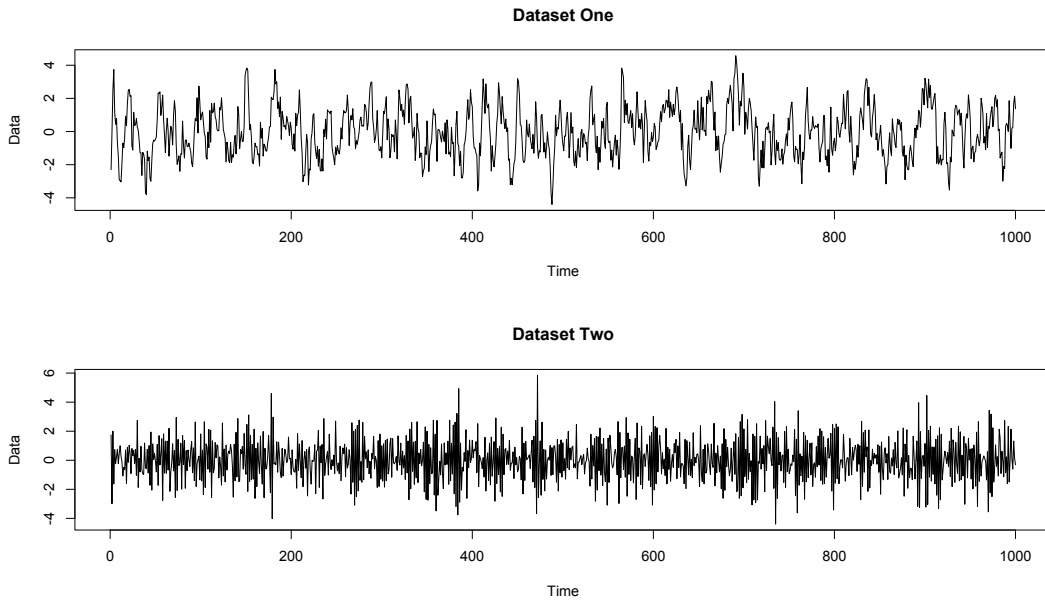


Figure 1: Two time-series datasets

5. **DFT of a periodic series.**

- a) Suppose $y_0, y_1, \dots, y_{n-1}$ is periodic with period h , meaning that

$$y_{t+hu} = y_t$$

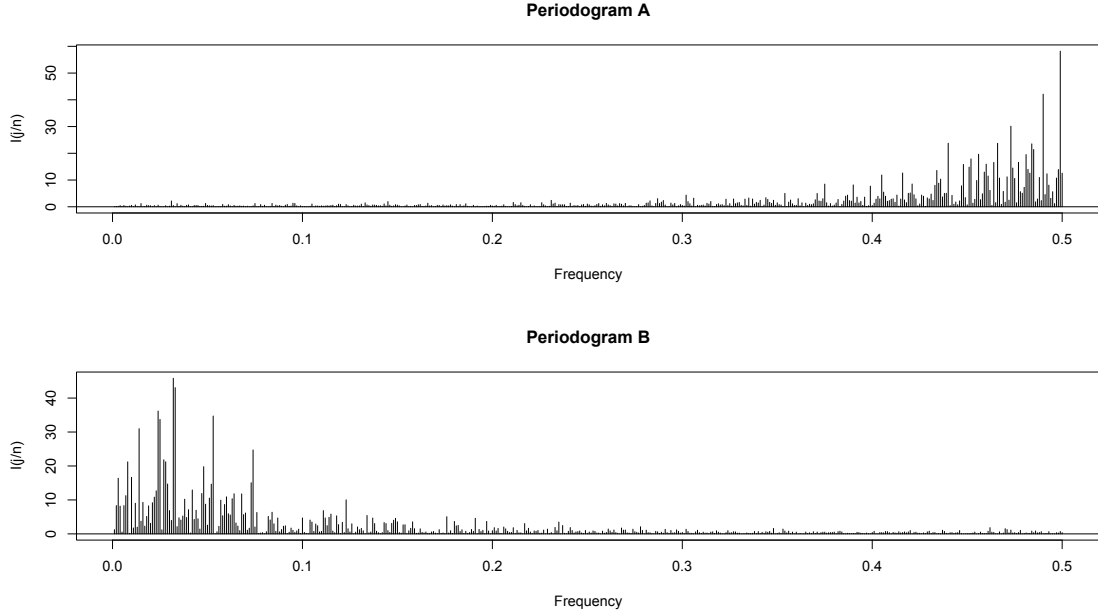


Figure 2: Two periodograms

for all integers t and u . Let n be an integer multiple of h , so that $n = kh$ for some integer k .

Suppose that the DFT of $y_0, \dots, y_{n-1}$ is $b_0, b_1, \dots, b_{n-1}$, and that the DFT of the data in the first cycle,

$$y_0, y_1, \dots, y_{h-1},$$

is $\beta_0, \dots, \beta_{h-1}$.

Show that

$$b_0 = k\beta_0, \quad b_k = k\beta_1, \quad b_{2k} = k\beta_2, \quad \dots, \quad b_{(h-1)k} = k\beta_{h-1},$$

and that all other b_j 's are zero. **(7 points)**

- b) Download the Google Trends data for the United States for the query *halloween*. Use a monthly time series beginning in January 2004. Remove the final two months so that the resulting dataset has length 252, corresponding to January 2004 through December 2024.

Plot the data and describe its main characteristics. **(2 points)**

- c) Plot the periodogram of this data. Explain how it aligns with the mathematical fact derived in Part (a). **(5 points)**

6. Download the Google Trends data for the United States for the query *golf*. Use a monthly time series beginning in January 2004 and ending in August 2026.

Fit the model

$$Y_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 \cos(2\pi f t) + \beta_4 \sin(2\pi f t) + \epsilon_t, \quad (3)$$

where

$$\epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

- a) Provide a point estimate and a 95% uncertainty interval for the unknown frequency parameter f . **(5 points)**
- b) On a scatter plot of the data, plot your best estimate of the fitted function

$$t \mapsto \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 \cos(2\pi f t) + \beta_4 \sin(2\pi f t)$$

along with appropriate uncertainty quantification. **(5 points)**

- c) Comment on whether model (3) is appropriate for this dataset. **(2 points)**

7. Download the FRED dataset on Total Construction Spending in the United States from

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This is monthly seasonally adjusted data on total construction spending in the United States. Fit the model

$$Y_t = \beta_0 + \beta_1 t + \beta_2(t - s_1)_+ + \beta_3(t - s_2)_+ + \epsilon_t, \quad (4)$$

where

$$\epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2), \quad (x)_+ := \max\{x, 0\}.$$

The unknown parameters are

$$\beta_0, \beta_1, \beta_2, \beta_3, s_1, s_2, \sigma.$$

1. Provide point estimates for the change-of-slope parameters s_1 and s_2 . **(5 points)**
2. On a scatter plot of the data, plot your best estimate of the fitted function

$$t \mapsto \beta_0 + \beta_1 t + \beta_2(t - s_1)_+ + \beta_3(t - s_2)_+.$$

(4 points)

3. Comment on whether model (4) is appropriate for this dataset. **(3 points)**

For two datasets

$$x_0, x_1, \dots, x_{n-1} \quad \text{and} \quad y_0, y_1, \dots, y_{n-1},$$

their *convolution* is the dataset $z_0, z_1, \dots, z_{n-1}$ defined by

$$z_i = \sum_{j=0}^{n-1} x_{i-j} y_j, \quad i = 0, \dots, n-1,$$

where indices are interpreted periodically, so that $x_{-k} = x_{-k+n}$.

Find the DFT of $z_0, \dots, z_{n-1}$ in terms of the DFTs of $x_0, \dots, x_{n-1}$ and $y_0, \dots, y_{n-1}$. **(8 points)**

The file `DataProblem9HomeworkTwo248Fall12025.csv` contains a time-series dataset with two columns. The first column is the time index t ; time is not equally spaced. The second column is the value y_t .

The data were generated according to the model

$$y_i = \beta_0 + \beta_1 \cos(2\pi f_1 t_i) + \beta_2 \sin(2\pi f_1 t_i) \\ + \beta_3 \cos(2\pi f_2 t_i) + \beta_4 \sin(2\pi f_2 t_i) + \epsilon_i,$$

where

$$\epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

1. Provide point estimates and 95% uncertainty intervals for f_1 and f_2 . (**5 points**)
2. Provide point estimates and 95% uncertainty intervals for

$$\beta_0, \beta_1, \beta_2, \beta_3, \beta_4,$$

as well as for σ . (**9 points**)